BCSC 0806: MICROPROCESSORS LAB

Objective: *The objective is to introduce the Architecture and programming of the microprocessor and learning about interfacing and various applications of microprocessor.*

Credits: 01

L–T–P-J: 0–0–2-0

Module No.	Content	Lab Hours
I and II	• To study 8085 microprocessor System. • Write an assembly language program to add two 8-bit numbers • Write an assembly language program to Subtract two 8-bit numbers • Write an assembly language program to Add contents of two memory locations • Write an assembly language program to Add two 16-bit numbers • Write an assembly language program to Subtract two 16-bit numbers • Write an assembly language program to Finding one's complement of a 8-bit number and 16-bit number • Write an assembly language program to Finding Two's complement of a 8-bit number and 16-bit number • Write an assembly language program to Pack the unpacked BCD numbers • Write an assembly language program to Unpack a BCD number • Write an assembly language program to Right shift, bit of data(8-bit and 16-bit) • Write an assembly language program to Left Shifting of data(8-bit and 16-bit) • Write an assembly language program to check the number is odd or even. • Write an assembly language program to check the number is positive or negative. • Write an assembly language program to Count Number of one's in a number • Write an assembly language program to Arrange in ascending order • Write an assembly language program to calculate the sum of series • Write an assembly language program to Calculate the sum of series of even numbers • Write an assembly language program to Calculate the sum of series of odd numbers • Write an assembly language program to calculate the sum of series of positive numbers. • Write an assembly language program to calculate the sum of series of negative numbers. • Write an assembly language program to Count the even numbers from a series.	24

	• **Write an assembly language program to Count the odd numbers from a series.** • **Write an assembly language program to Count the negative numbers from a series.** • **Write an assembly language program to Count the positive numbers from a series.** • **Write an assembly language program to Add two decimal numbers** • **Write an assembly language program to separate even numbers from given numbers** • **Write an assembly language program to Transfer contents from memory block B1 to memory block B2.**	

Reference Books:

- Ramesh S. Gaonkar (2000), "*Microprocessor Architecture Programming and Applications with 8085*", 4th Edition, Penram International Publishing.
- D. V. Hall (1992), "*Microprocessors and Interfacing: Programming and Hardware*", 2nd Edition, TMH.

Outcome: After completion of course, student will be able to:

- Understand the concepts of advanced microprocessors.
- Understand various interfacing circuits necessary for various applications.